Hepatitis B Core Antibody, Total

Test Details

METHODOLOGY

Immunochemiluminometric assay (ICMA)

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This assay can be used as an aid in the diagnosis of individuals with acute or chronic hepatitis B virus (HBV) infection and in the determination of the clinical status of HBV infected individuals in conjunction with other HBV serological markers associated with HBV infection. This assay can also be used as an aid in the differential diagnosis in individuals displaying signs of hepatitis in whom etiology is unknown.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

The results from this or any other diagnostic kit should be used and interpreted only in the context of the overall clinical picture. A negative test result does not exclude the possibility of exposure to hepatitis B virus. Levels of anti-HBc may be undetectable both in early infection and later after infection.

This is a qualitative assay; reported antibody level cannot be correlated to an endpoint titer.

Assay performance characteristics have not been established for immunocompromised or immunosuppressed patients.

Custom Additional Information

Hepatitis B core antigen (HBcAg), found in liver cells, does not circulate in the bloodstream. However, IgM and IgG antibodies to HBcAg can be detected serologically in HBV infected individuals. Anti-HBc IgM is detectable first and remains detectable for approximately six months. Shortly after the IgM response, anti-HBc IgG appears and can remain detectable indefinitely. The presence of anti-HBc IgM and anti-HBc IgG is characteristic of acute infection, while the presence of anti-HBc IgG without anti-HBc IgM is characteristic of chronic or recovered stages of HBV infection. Anti-HBc Total assays detect both IgM and IgG anti-HBc responses. Most often levels of anti-HBc will coincide with detectable levels of other HBV markers. Rarely, anti-HBc may be the only detectable HBV marker. This may occur during the brief period when hepatitis B surface antigen (HBsAg) has been cleared from the bloodstream and before antibodies to hepatitis B surface antigen (anti-HBs) become detectable. For this reason, the use of anti-HBc Total assays to detect acute infection is not recommended. Anti-HBc Total assays should be used in conjunction with other marker assays to assess current or past exposure to HBV.

Specimen Requirements

Specimen

Serum **or** plasma

Volume

1 mL

Minimum Volume

0.4 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** lavender-top (EDTA) tube

Collection Instructions

If tube other than a gel-barrier tube is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	7 days (stability provided by manufacturer or literature reference)
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

Negative

Storage Instructions

Room temperature

Causes for Rejection

Non-EDTA plasma specimen; PST gel-barrier tube

References

Abara WE, Qaseem A, Schillie S, et al. Hepatitis B Vaccination, Screening, and Likage to Care: Best Practice Advice from American College of Physicians and the Centers for Disease Control and Prevention. *Ann Intern Med.* 2017 Dec 5;167(11):794-804. PubMed 29159414 (http://www.ncbi.nlm.nih.gov/pubmed/29159414)

Schillie S, Vellozzi C, Reingold A, et al. Prevention of Hepatitis B Virus Infection in the United States. Recommendations of the Advisory Committee on Immunization Practices. *MMWR Recomm Rep.* 2018 Jan 12;67(1):1-31. PubMed 29939980 (http://www.ncbi.nlm.nih.gov/pubmed/29939980)

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